

Notes: Kamenica & Gentzkow (AER, 2011)

Bayesian Persuasion

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1 Big picture

- **Question.** Can a sender persuade a rational Bayesian receiver by choosing what information structure the receiver observes, even when both players understand the sender's incentives?
- **Core answer.** Yes. Bayes' rule restricts the *average* posterior to equal the prior, but it does not otherwise restrict the distribution of posteriors. A sender can exploit nonlinearities in the receiver's action rule.
- **Canonical example.** A prosecutor with prior guilt probability 0.3 can choose an investigation that makes a rational judge convict with probability 0.6, even though full disclosure would convict with probability only 0.3.
- **Main conceptual move.** Replace the problem of choosing a signal with the problem of choosing a Bayes-plausible distribution of posterior beliefs.
- **Geometric solution.** Let $\hat{v}(\mu)$ be the sender's payoff if the receiver holds posterior μ and best responds. The sender's optimal value at prior μ_0 is the concave closure $V(\mu_0)$.
- **When persuasion helps.** Persuasion helps if $V(\mu_0) > \hat{v}(\mu_0)$. Intuitively, the sender needs some belief at which the receiver takes a better action, and some offsetting belief that does not hurt the sender too much.
- **Optimal signals.** An optimal signal induces posterior beliefs at contact points between $\hat{v}$ and its concave closure. In binary-state examples, this can be read directly from a graph.
- **Economic mechanisms.** The optimal experiment often separates receivers into a favorable group just willing to take the sender's preferred action and an unfavorable group whose action would not have helped the sender anyway.
- **Applications.** The framework covers courtroom evidence, lobbying, advertising, consumer trials, political communication, financial disclosure, and any environment where an informed party can commit to a signal before information is realized.

2 Incremental contribution of the paper

- **From cheap talk to information design.** The paper shifts the classic strategic communication question from "what message can a biased sender credibly send?" to "what experiment would a biased sender choose if truthful reporting of the experiment's outcome is guaranteed?"
- **Concavification as a general tool.** It imports and popularizes the concavification method for persuasion problems. After the paper, many information-design papers use the same three-step logic: compute the indirect payoff, concavify it, and implement the supporting distribution of posteriors.
- **Bayes plausibility as the only feasibility restriction.** Proposition 1 shows that a distribution of posteriors is implementable if and only if its mean is the prior. This creates an extremely simple feasibility condition.
- **Sharp characterization of optimal persuasion.** The paper does not just show that persuasion is possible; it characterizes the sender's optimal signal and provides necessary and sufficient conditions for gains from persuasion.
- **General finite-state/action framework.** Prior work studied narrower environments, such as binary states or parameterized disclosure policies. This paper handles arbitrary finite states, compact actions, arbitrary priors, and state-dependent preferences.
- **New economic intuition.** Persuasion can be effective even with fully rational receivers because the receiver's action is nonlinear in beliefs. The sender manipulates the distribution of beliefs without changing the average belief.

- **Application discipline.** The court, lobbying, and advertising examples show how the same abstract posterior-splitting problem can interpret seemingly different economic institutions.
- **Foundation for later work.** The paper establishes the benchmark single-receiver information-design model against which later work studies multiple receivers, privacy constraints, persuasion with learning, robust persuasion, limited attention, and dynamic persuasion.

3 Model primitives and measurement

3.1 Players, states, actions, and payoffs

- There is a finite state space Ω and compact receiver action space A .
- Sender and Receiver share a common prior $\mu_0 \in \text{int}(\Delta(\Omega))$.
- Receiver's utility is $u(a, \omega)$; Sender's utility is $v(a, \omega)$.
- Sender chooses a signal $\pi : S \rightarrow \Delta(\Omega)$, or equivalently conditional probabilities $\pi(s \mid \omega)$.
- Receiver observes the signal realization s , updates to posterior μ_s , and chooses an action.
- Sender cannot distort the realized signal; the commitment is to the information structure, not to ex post cheap-talk messages.

Interpretation. The model captures settings where a sender can commit to an experiment, investigation, rating scale, product trial, or disclosure rule, and then truthfully reveal the realized outcome.

3.2 Receiver's best response and sender's indirect payoff

Receiver chooses an action that maximizes expected utility:

$$\hat{a}(\mu) \in \arg \max_{a \in A} E_\mu[u(a, \omega)].$$

Taking the receiver's behavior as given, the sender's indirect payoff at belief μ is

$$\hat{v}(\mu) \equiv E_\mu[v(\hat{a}(\mu), \omega)].$$

Interpretation. The object $\hat{v}$ summarizes everything the sender cares about once the receiver has a posterior. The persuasion problem becomes a problem of moving the receiver across different posteriors.

3.3 Posterior distributions and Bayes plausibility

A signal induces a distribution $\tau \in \Delta(\Delta(\Omega))$ over posterior beliefs. The key feasibility condition is

$$\sum_{\mu \in \text{Supp}(\tau)} \mu \tau(\mu) = \mu_0.$$

Interpretation. This is the martingale property of Bayesian posteriors: posterior beliefs can spread out, but they must average back to the prior. This is why persuasion works through changing the *dispersion* of beliefs, not their mean.

3.4 Value of a signal

If a signal induces posterior distribution τ , Sender's value is

$$E_\tau[\hat{v}(\mu)].$$

Sender benefits from persuasion if there is a Bayes-plausible τ such that

$$E_\tau[\hat{v}(\mu)] > \hat{v}(\mu_0).$$

Interpretation. The sender wants to create a mean-preserving spread of beliefs that raises the expectation of $\hat{v}$. Whether this is possible depends on curvature and kinks in $\hat{v}$.

3.5 Concave closure

Define the concave closure V of $\hat{v}$ as

$$V(\mu) = \sup\{z \mid (\mu, z) \in \text{co}(\hat{v})\},$$

where $\text{co}(\hat{v})$ is the convex hull of the graph of $\hat{v}$.

Interpretation. The concave closure is the smallest concave function weakly above $\hat{v}$. Its value at μ_0 is the sender's maximum attainable expected payoff.

4 Models and theoretical specifications

4.1 Proposition 1: Bayes plausibility and the revelation-principle logic

Proposition 1 states that the following are equivalent:

1. there exists a signal with value v^* ;
2. there exists a straightforward signal with value v^* ;
3. there exists a Bayes-plausible distribution of posteriors τ such that $E_\tau \hat{v}(\mu) = v^*$.

Interpretation. A signal can be replaced by a straightforward recommendation signal that induces the same receiver actions. Therefore, the sender only needs to choose a feasible distribution of posteriors.

4.2 Corollary 2: optimal value and strict gain from persuasion

The value of an optimal signal is

$$V(\mu_0).$$

Sender benefits from persuasion if and only if

$$V(\mu_0) > \hat{v}(\mu_0).$$

Interpretation. This is the core formula of the paper. The entire persuasion problem is reduced to comparing the indirect payoff function with its concave closure at the prior.

4.3 Motivating prosecutor example

The judge convicts if posterior guilt probability is at least 0.5. The prior is $\mu_0 = 0.3$. The prosecutor's optimal binary investigation is

$$\begin{aligned}\pi(i \mid \textit{innocent}) &= \frac{4}{7}, & \pi(i \mid \textit{guilty}) &= 0, \\ \pi(g \mid \textit{innocent}) &= \frac{3}{7}, & \pi(g \mid \textit{guilty}) &= 1.\end{aligned}$$

The signal i induces posterior 0, and signal g induces posterior 0.5; the judge convicts after g and acquits after i . The conviction probability is 0.6.

Interpretation. The judge is rational and understands the prosecutor's incentives. Persuasion works because unfavorable information is pooled into a signal that merely confirms acquittal, while favorable information just reaches the conviction threshold.

4.4 When does persuasion help?

The paper defines that there is information Sender would share if there is a belief μ such that

$$\hat{v}(\mu) > E_\mu[v(\hat{a}(\mu_0), \omega)].$$

Proposition 2 says:

- if there is no information Sender would share, persuasion cannot help;
- if there is information Sender would share and Receiver's preference is discrete at the prior, persuasion helps;
- if A is finite, discreteness at the prior is generic.

Interpretation. The sender needs one posterior that would induce a better action and another nearby posterior that leaves the default action unchanged. This generates a favorable mean-preserving spread.

4.5 Expected-state simplification

When Sender's payoff depends only on $E_\mu[\omega]$, define $\tilde{v}$ and its concave closure $\tilde{V}$. Proposition 3 says Sender benefits from persuasion if and only if

$$\tilde{V}(E_{\mu_0}[\omega]) > \tilde{v}(E_{\mu_0}[\omega]).$$

Interpretation. This reduces a high-dimensional belief problem to a one-dimensional graph over the expected state. It is especially useful in advertising and consumer-learning applications.

4.6 Properties of optimal posteriors

Proposition 4 states that if an optimal signal induces a belief leading to a worst action for Sender, then Receiver is certain of her action at that belief. Proposition 5 states that under Assumption 1 and positive persuasion value, if an optimal posterior is interior or leads to a best-attainable action, Receiver's preference is not discrete there.

Interpretation. These results describe the logic of optimal experiments. Bad outcomes are made as bad and conclusive as possible, while good outcomes often place the receiver exactly at an indifference threshold.

4.7 Extensions

If Receiver has private information r , Sender's indirect payoff becomes

$$\hat{v}(\mu) = \sum_{\omega \in \Omega} \sum_{r \in R} v(\hat{a}(\mu, r), \omega) \chi(r \mid \omega) \mu(\omega).$$

The concavification approach still applies after this redefinition. With multiple receivers, the simple reduction can fail because a receiver's action at a given belief may depend on beliefs about others' signals.

Interpretation. The single-receiver benchmark is robust to private information but not automatically to strategic interactions among multiple receivers.

5 Identification and theoretical design

5.1 What makes the argument work

- **Commitment.** Sender commits to a signal before the state is learned or before the signal realization is observed.
- **Truthful revelation of realizations.** Once a signal is chosen, its realized message is observed by Receiver; Sender cannot hide or misreport it.
- **Rational Bayesian updating.** Receiver correctly uses Bayes' rule and knows the signal structure.
- **No transfers or direct payoff manipulation.** Sender influences Receiver only by information.
- **Bayes plausibility.** The induced posterior distribution must average to the prior; Proposition 1 shows this is also sufficient.

Interpretation. The paper isolates persuasion through endogenous information structures. There is no bounded rationality, deception, transfer, or enforcement constraint doing the work.

5.2 Limits of the framework

- The sender must commit to the information structure.
- The receiver must observe the realized message and understand the experiment.
- The benchmark model has one receiver; with multiple strategic receivers, posterior distributions alone may not determine actions.
- The model characterizes optimal information design, not how commitment institutions arise.
- The paper is theoretical and does not estimate the magnitude of persuasion in real markets or politics.

Interpretation. These limitations clarify the right empirical applications: experiments, ratings, trials, investigations, disclosure rules, and institutional designs where commitment to a signal is plausible.

6 Figures and extracted proposition blocks

6.1 Figure 1: An Illustration of Concave Closure and Proposition 1: Bayes-Plausible Posterior Distributions

Read:

- Figure 1 plots $\hat{v}(\mu)$ in black and the concave closure $V(\mu)$ in red.
- The shaded region is the convex hull of the graph of $\hat{v}$; the upper boundary is V .
- If the prior lies between μ_l and μ_h , the sender can induce a two-posterior signal supported at those two beliefs.
- Proposition 1 states that this distribution of posteriors is implementable exactly when it is Bayes plausible.
- Together, the figure and proposition show why solving for an optimal signal is equivalent to finding a supporting chord of the concave closure.

Economic message. The graph turns information design into geometry: sender value is the height of the concave envelope at the prior, and the optimal signal corresponds to the endpoints of the chord supporting that point.

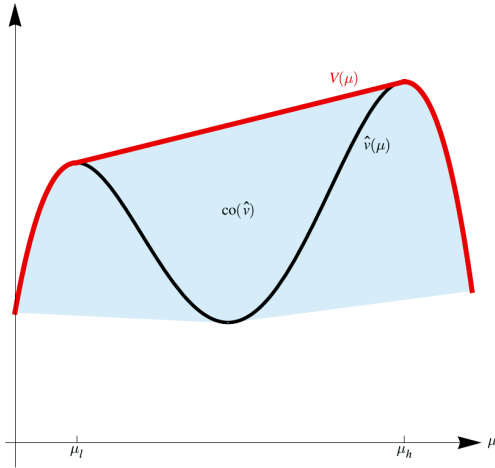


FIGURE 1. AN ILLUSTRATION OF CONCAVE CLOSURE

(a) Figure 1: concave closure of $\hat{v}$

of our game. We simplify the analysis further by noting that, without loss of generality, we can restrict our attention to a particular class of signals. Say that a signal is *straightforward* if $S \subseteq A$ and Receiver's equilibrium action equals the signal realization. In other words, a straightforward signal produces a "recommended action" and Receiver always follows the recommendation.

PROPOSITION 1: *The following are equivalent:*

- (i) *There exists a signal with value v^* ;*
- (ii) *There exists a straightforward signal with value v^* ;*
- (iii) *There exists a Bayes-plausible distribution of posteriors τ such that $E_\tau \hat{v}(\mu) = v^*$.*

(b) Proposition 1: Bayes-plausible posterior distributions

6.2 Corollary 2: The Value of an Optimal Signal and Figure 2: The Motivating Example

Read:

- Corollary 2 states that the optimal value is $V(\mu_0)$ and persuasion is valuable exactly when $V(\mu_0) > \hat{v}(\mu_0)$.
- In Figure 2, Panel A shows the prosecutor's payoff as a step function: zero below the conviction threshold and one at or above it.
- Panel B shows the optimal posterior split: one posterior at zero and another at the conviction threshold 0.5.
- Panel C shows the concave closure, which lies above the step function for priors below 0.5.
- With prior 0.3, the prosecutor raises conviction probability from zero under no information to 0.6 under the optimal signal.

Economic message. The sender does not need to make the receiver overly optimistic. The optimal persuasion signal makes the receiver just indifferent at the favorable signal and certain enough to maintain the default action at the unfavorable signal.

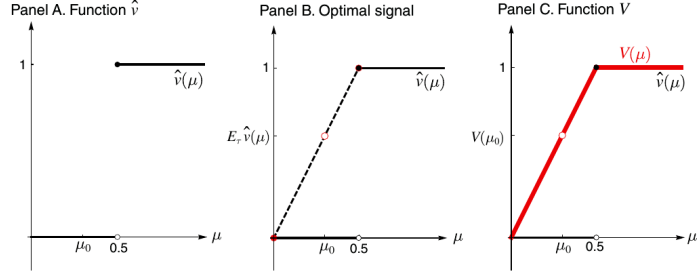


FIGURE 2. THE MOTIVATING EXAMPLE

COROLLARY 2: The value of an optimal signal is $V(\mu_0)$. Sender benefits from persuasion if and only if $V(\mu_0) > \hat{v}(\mu_0)$.

(a) Corollary 2: value of optimal signal

(b) Figure 2: motivating prosecutor example

6.3 Proposition 2: Conditions for Sender to Benefit and Proposition 3: Expected-State Simplification

Read:

- Proposition 2 gives primitive conditions for persuasion to be valuable: there must be information the sender would share, and the receiver's preference must be locally stable at the prior.
- This proposition explains why finite-action environments are naturally persuasion-friendly: exact indifference is non-generic, so actions are usually locally constant.
- Proposition 3 shows that when the sender's payoff depends only on the expected state, the high-dimensional concavification problem collapses to a one-dimensional one.
- This is the key technical simplification used for applications such as advertising and lobbying.

Economic message. Persuasion is strongest when a small amount of good news can trigger a discrete action change, while a compensating amount of bad news does not change the receiver's default action.

PROPOSITION 2: If there is no information Sender would share, Sender does not benefit from persuasion. If there is information Sender would share and Receiver's preference is discrete at the prior, Sender benefits from persuasion. If A is finite, Receiver's preference is discrete at the prior generically.

(a) Proposition 2: conditions for persuasion to help

PROPOSITION 3: When Sender's payoff depends only on the expected state, Sender benefits from persuasion if and only if $\bar{V}(E_{\mu_0}[\omega]) > \hat{v}(E_{\mu_0}[\omega])$.

(b) Proposition 3: expected-state representation

6.4 Propositions 4 and 5: Structure of Optimal Posteriors

Read:

- Proposition 4 says that if the receiver chooses the sender's worst action after an optimal signal, the receiver must be certain of that action.
- Proposition 5 says that favorable or interior posteriors induced by an optimal signal typically leave the receiver just indifferent between actions.
- These results formalize the intuition from the prosecutor example: acquittal posteriors put no weight on guilty defendants, while conviction posteriors put the judge exactly at the conviction threshold.
- The proposition block also includes Assumption 1, which rules out pathological payoff ties that would otherwise break the indifference characterization.

Economic message. Optimal persuasion economizes on good news. It does not waste favorable realizations by making the receiver strictly prefer the sender’s desired action; instead, it pushes her only to the margin where she is willing to act.

PROPOSITION 4: *If an optimal signal induces a belief μ that leads to a worst action, Receiver is certain of her action at μ .*

The second of the properties above, on the other hand, is not completely general when v depends on ω . Sender’s preferences could then take a form such that two actions a_1 and a_2 are both induced by an optimal signal, but at the belief at which Receiver takes a_1 Sender is exactly indifferent between a_1 and a_2 . Then the logic described in the introduction—where Sender could improve his payoff by shifting probability mass away from a belief that induced a strict preference for one action in order to increase the probability of an alternative action—need not hold, since Sender is exactly indifferent to this change. However, if this occurs under a particular utility function v , it will not occur for a slight perturbation of v . The following assumption rules out this pathological case.

ASSUMPTION 1: *There exists no action a s.t. (i) $\forall \mu, \hat{v}(\mu) \leq E_\mu v(a, \omega)$ and (ii) $\exists \mu$ s.t. $a \neq \hat{a}(\mu)$ and $\hat{v}(\mu) = E_\mu v(a, \omega)$.*

The intuition that we provided for property (ii) applies to any interior belief. The same intuition, however, also applies to any belief which leads to an action that Sender cannot improve upon. Say an action a is *best attainable* if $E_\mu v(a, \omega) > \hat{v}(\mu)$ for all μ s.t. $a \neq \hat{a}(\mu)$.

PROPOSITION 5: *Suppose Assumption 1 holds and Sender benefits from persuasion. If a belief μ induced by an optimal signal is either (i) interior or (ii) leads to a best-attainable action, then Receiver’s preference is not discrete at μ .*

Propositions 4 and 5: restrictions on beliefs induced by an optimal signal

6.5 Figure 3: Advertising to Increase Sales and the Extension with Receiver Private Information

Read:

- Figure 3 applies the expected-state representation to advertising or product trials.
- A consumer buys if expected match value exceeds the outside option threshold $\underline{u}$.
- The optimal signal splits consumers into a group that is sure enough not to buy and a group just willing to buy.
- The private-information extension shows that the same method applies if Receiver has her own private signal; the only change is the definition of $\hat{v}(\mu)$.
- Multiple receivers are harder because a receiver’s action may depend on beliefs about what information other receivers receive.

Economic message. In advertising, the optimal trial can be honest and selective: it convinces consumers for whom the product is just good enough and separates out consumers who should not buy. This can increase sales without misleading rational consumers.

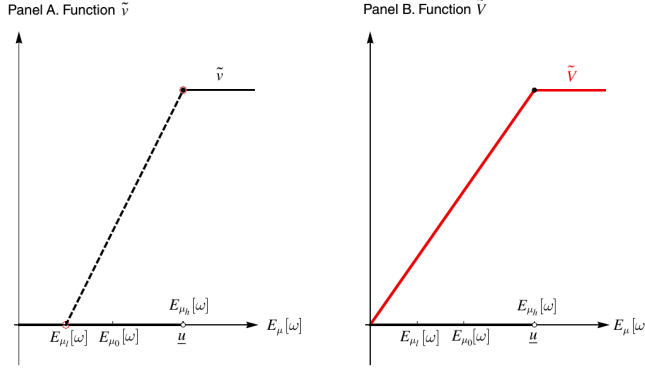


FIGURE 3. ADVERTISING TO INCREASE SALES

(a) Figure 3: advertising to increase sales

The only way in which Receiver's private information changes our analysis is that we can no longer construct a deterministic function $\hat{d}(\mu)$ which specifies Receiver's action at any belief she shares with Sender. Rather, for any Sender's belief μ , Receiver's optimal action $\hat{d}(\mu, r)$ depends on the realization of her private signal and so is stochastic from Sender's perspective. When his posterior is μ , Sender assigns probability $\chi(r|\omega)\mu(\omega)$ to the event that Receiver's signal is r and the state is ω . Hence, Sender's expected utility when his posterior is μ is:

$$(3) \quad \hat{v}(\mu) = \sum_{\omega \in \Omega} \sum_{r \in R} v(\hat{d}(\mu, r), \omega) \chi(r|\omega) \mu(\omega).$$

(b) Extension: receiver private information

7 Short summary

- The paper studies persuasion by information design under full Bayesian rationality.
- Sender commits to a signal; Receiver observes the realized signal and best responds.
- Bayes plausibility is the only restriction on the distribution of posterior beliefs.
- The sender's problem is to maximize $E_{\tau} \hat{v}(\mu)$ over Bayes-plausible posterior distributions.
- The optimal value is the concave closure $V(\mu_0)$ of the sender's indirect payoff at the prior.
- Persuasion is valuable exactly when $V(\mu_0) > \hat{v}(\mu_0)$.
- In the prosecutor example, the optimal signal increases conviction probability from zero under no information to 0.6.
- The optimal signal often induces posteriors at action thresholds rather than extreme favorable beliefs.
- If $\hat{v}$ is concave, no disclosure is optimal; if it is convex, full disclosure is optimal.
- If Sender's payoff depends only on the expected state, the problem can be reduced to a one-dimensional concavification.
- The framework applies to lobbying, advertising, litigation, disclosure, product trials, and many other environments.

8 Conclusion

8.1 Core conclusion

Kamenica and Gentzkow show that a biased sender can persuade a fully rational Bayesian receiver by designing the receiver's information environment. The key is not deception. The receiver knows the sender's preferences and applies Bayes' rule correctly. Persuasion works because Bayes' rule fixes the mean posterior but not the distribution of posteriors, while the receiver's action is usually nonlinear in beliefs.

8.2 Economic mechanism

The sender creates dispersion in beliefs. Favorable signals push the receiver across an action threshold; unfavorable signals compensate for Bayes plausibility but often do not hurt the sender because the receiver

would have taken the default action anyway. The sender’s value is therefore an envelope problem: choose posterior beliefs that average to the prior and maximize the expected indirect payoff.

8.3 Incremental contribution for the literature

The paper’s incremental contribution is to transform strategic persuasion into a tractable information-design problem. Earlier communication models emphasized incentive compatibility, cheap talk, or fixed disclosure rules. This paper instead characterizes the optimal information structure itself and shows that the feasibility constraint is simply Bayes plausibility. The concavification method became the organizing tool for the subsequent Bayesian persuasion literature.

8.4 Practical interpretation

In a courtroom, the optimal investigation need not reveal all facts; it may optimally group evidence into categories that either make conviction just justified or make acquittal certain. In advertising, a seller may optimally design product trials that screen out consumers for whom the product is poor while making marginal consumers just willing to buy. In lobbying, an interest group may fund studies that create the right distribution of posterior beliefs among policymakers.

8.5 Limits and open questions

The benchmark requires commitment to the experiment and truthful revelation of realized messages. It also relies on a single receiver whose action is determined by her posterior. Later research relaxes these assumptions by studying multiple receivers, privacy constraints, dynamics, ambiguity, robustness, costly attention, and limited commitment. The paper remains the canonical single-receiver benchmark against which these extensions are measured.

8.6 Takeaway

The central takeaway is: rationality does not eliminate persuasion. It disciplines persuasion. A rational receiver cannot be made to hold biased average beliefs, but she can be exposed to a carefully chosen distribution of beliefs. The sender’s problem is to choose that distribution.